DATABASE RELATIONSHIP ANALYSIS USING JOINS – PURPLE E-COMMERCE MANAGEMENT SYSTEM

1. Create the tables

```
SQL> CREATE TABLE PURPLE_CUSTOMER (
  2       CUSTOMER_ID NUMBER PRIMARY KEY,
  3       CUSTOMER_NAME VARCHAR2(50),
  4       CITY VARCHAR2(30)
  5  );

Table created.

SQL>
SQL> CREATE TABLE PURPLE_PRODUCT (
  2       PRODUCT_ID NUMBER PRIMARY KEY,
  3       PRODUCT_NAME VARCHAR2(50),
  4       PRICE NUMBER(10,2)
  5  );

Table created.

SQL>
SQL> CREATE TABLE PURPLE_ORDER (
  2       ORDER_ID NUMBER PRIMARY KEY,
  3       CUSTOMER_ID NUMBER REFERENCES PURPLE_CUSTOMER(CUSTOMER_ID),
  4       ORDER_DATE DATE,
  5       TOTAL_AMOUNT NUMBER(10,2)
  6  );

Table created.

SQL>
SQL> CREATE TABLE PURPLE_ORDER_ITEM (
  2       ORDER_ITEM_ID NUMBER PRIMARY KEY,
  3       ORDER_ID NUMBER REFERENCES PURPLE_ORDER(ORDER_ID),
  4       PRODUCT_ID NUMBER REFERENCES PURPLE_PRODUCT(PRODUCT_ID),
  5       QUANTITY NUMBER,
  6       PRICE NUMBER(10,2)
  7  );

Table created.
```

```
SQL> CREATE TABLE PURPLE_PAYMENT (
  2       PAYMENT_ID NUMBER PRIMARY KEY,
  3       ORDER_ID NUMBER REFERENCES PURPLE_ORDER(ORDER_ID),
  4       PAYMENT_METHOD VARCHAR2(30),
  5       PAYMENT_STATUS VARCHAR2(20),
  6       PAYMENT_DATE DATE,
  7       AMOUNT NUMBER(10,2)
  8  );

Table created.

SQL>
SQL> CREATE TABLE PURPLE_CATEGORY (
  2       CATEGORY_ID NUMBER PRIMARY KEY,
  3       CATEGORY_NAME VARCHAR2(50),
  4       DESCRIPTION VARCHAR2(100)
  5  );

Table created.

SQL>
SQL> CREATE TABLE PURPLE_SUPPLIER (
  2       SUPPLIER_ID NUMBER PRIMARY KEY,
  3       SUPPLIER_NAME VARCHAR2(50),
  4       CITY VARCHAR2(30),
  5       PHONE NUMBER(10)
  6  );

Table created.

SQL>
SQL> CREATE TABLE PURPLE_INVENTORY (
  2       INVENTORY_ID NUMBER PRIMARY KEY,
  3       PRODUCT_ID NUMBER REFERENCES PURPLE_PRODUCT(PRODUCT_ID),
  4       STOCK_QUANTITY NUMBER,
  5       LAST_UPDATED DATE
  6  );

Table created.
```

```
SQL>
SQL> CREATE TABLE PURPLE_REVIEW (
  2      REVIEW_ID NUMBER PRIMARY KEY,
  3      CUSTOMER_ID NUMBER REFERENCES PURPLE_CUSTOMER(CUSTOMER_ID),
  4      PRODUCT_ID NUMBER REFERENCES PURPLE_PRODUCT(PRODUCT_ID),
  5      RATING NUMBER(1),
  6      REVIEW_TEXT VARCHAR2(200),
  7      REVIEW_DATE DATE
  8  );

Table created.
```

1.1Insert sample data

```
SQL> -- CUSTOMER
SQL> INSERT INTO PURPLE_CUSTOMER VALUES (101, 'Riya', 'Chennai');

1 row created.

SQL> INSERT INTO PURPLE_CUSTOMER VALUES (102, 'Bhava', 'Chrompet');

1 row created.

SQL> INSERT INTO PURPLE_CUSTOMER VALUES (103, 'Sudha', 'Tambaram');

1 row created.

SQL>
SQL> -- PRODUCT
SQL> INSERT INTO PURPLE_PRODUCT VALUES (201, 'Keyboard', 999);

1 row created.

SQL> INSERT INTO PURPLE_PRODUCT VALUES (202, 'Mouse', 499);

1 row created.

SQL> INSERT INTO PURPLE_PRODUCT VALUES (203, 'Headphones', 1499);

1 row created.

SQL>
SQL> -- ORDERS
SQL> INSERT INTO PURPLE_ORDER VALUES (1001, 101, DATE '2026-10-20', 1498);

1 row created.

SQL> INSERT INTO PURPLE_ORDER VALUES (1002, 102, DATE '2026-10-21', 1499);

1 row created.

SQL> INSERT INTO PURPLE_ORDER VALUES (1003, 101, DATE '2026-10-22', 999);

1 row created.
```

```
SQL>
SQL> -- ORDER ITEMS
SQL> INSERT INTO PURPLE_ORDER_ITEM VALUES (1, 1001, 201, 1, 999);

1 row created.

SQL> INSERT INTO PURPLE_ORDER_ITEM VALUES (2, 1001, 202, 1, 499);

1 row created.

SQL> INSERT INTO PURPLE_ORDER_ITEM VALUES (3, 1002, 203, 1, 1499);

1 row created.

SQL> INSERT INTO PURPLE_ORDER_ITEM VALUES (4, 1003, 201, 1, 999);

1 row created.

SQL>
SQL> -- PAYMENTS
SQL> INSERT INTO PURPLE_PAYMENT VALUES
  2  (1, 1001, 'UPI', 'Successful', DATE '2026-10-20', 1498);

1 row created.

SQL>
SQL> INSERT INTO PURPLE_PAYMENT VALUES
  2  (2, 1002, 'Credit Card', 'Successful', DATE '2026-10-21', 1499);

1 row created.

SQL>
SQL> INSERT INTO PURPLE_PAYMENT VALUES
  2  (3, 1003, 'Debit Card', 'Successful', DATE '2026-10-22', 999);

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

1.2Display the table

```
SQL> SELECT * FROM PURPLE_CUSTOMER;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
CITY
--------------------------------
        101 Riya
Chennai

        102 Bhava
Chrompet

        103 Sudha
Tambaram


SQL> SELECT * FROM PURPLE_PRODUCT;

PRODUCT_ID PRODUCT_NAME                                            PRICE
---------- -------------------------------------------------- ----------
       201 Keyboard                                                  999
       202 Mouse                                                     499
       203 Headphones                                               1499

SQL> SELECT * FROM PURPLE_ORDER;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
---------- ----------- --------- ------------
      1001         101 20-OCT-26         1498
      1002         102 21-OCT-26         1499
      1003         101 22-OCT-26          999

SQL> SELECT * FROM PURPLE_ORDER_ITEM;

ORDER_ITEM_ID   ORDER_ID PRODUCT_ID   QUANTITY      PRICE
------------- ---------- ---------- ---------- ----------
            1       1001        201          1        999
            2       1001        202          1        499
            3       1002        203          1       1499
            4       1003        201          1        999
```

```
SQL> SELECT * FROM PURPLE_PAYMENT;

PAYMENT_ID   ORDER_ID PAYMENT_METHOD                 PAYMENT_STATUS
---------- ---------- ------------------------------ --------------------
PAYMENT_D     AMOUNT
--------- ----------
         1       1001 UPI                            Successful
20-OCT-26       1498

         2       1002 Credit Card                    Successful
21-OCT-26       1499

         3       1003 Debit Card                     Successful
22-OCT-26        999
```

2. Implement INNER JOIN, LEFT JOIN, and RIGHT JOIN

2.1 INNER JOIN

```
SQL> SELECT C.CUSTOMER_NAME,
  2         O.ORDER_ID,
  3         PR.PRODUCT_NAME,
  4         OI.QUANTITY,
  5         O.TOTAL_AMOUNT,
  6         P.PAYMENT_METHOD,
  7         P.PAYMENT_STATUS
  8  FROM PURPLE_CUSTOMER C
  9  INNER JOIN PURPLE_ORDER O
 10      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 11  INNER JOIN PURPLE_ORDER_ITEM OI
 12      ON O.ORDER_ID = OI.ORDER_ID
 13  INNER JOIN PURPLE_PRODUCT PR
 14      ON OI.PRODUCT_ID = PR.PRODUCT_ID
 15  INNER JOIN PURPLE_PAYMENT P
 16      ON O.ORDER_ID = P.ORDER_ID;

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME                                       QUANTITY TOTAL_AMOUNT
-------------------------------------------------- ---------- ------------
PAYMENT_METHOD                 PAYMENT_STATUS
------------------------------ --------------------
Riya                                                     1001
Keyboard                                                    1         1498
UPI                            Successful

Riya                                                     1003
Keyboard                                                    1          999
Debit Card                     Successful
```

2.2 LEFT JOIN

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         C.CITY,
  4         O.ORDER_ID,
  5         O.ORDER_DATE,
  6         O.TOTAL_AMOUNT
  7  FROM PURPLE_CUSTOMER C
  8  LEFT JOIN PURPLE_ORDER O
  9      ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
CITY                           ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
        101 Riya
Chennai                              1001 20-OCT-26         1498

        102 Bhava
Chrompet                             1002 21-OCT-26         1499

        101 Riya
Chennai                              1003 22-OCT-26          999


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
CITY                           ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
        103 Sudha
Tambaram
```

2.3 RIGHT JOIN

```
SQL> SELECT C.CUSTOMER_NAME,
  2         O.ORDER_ID,
  3         O.ORDER_DATE,
  4         O.TOTAL_AMOUNT
  5  FROM PURPLE_CUSTOMER C
  6  RIGHT JOIN PURPLE_ORDER O
  7      ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_NAME                                      ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
Riya                                                 1001 20-OCT-26
       1498

Riya                                                 1003 22-OCT-26
        999

Bhava                                                1002 21-OCT-26
       1499
```

3. Retrieve Complete Order Details

```
SQL> SELECT O.ORDER_ID,
  2           O.ORDER_DATE,
  3           C.CUSTOMER_NAME,
  4           C.CITY,
  5           PR.PRODUCT_NAME,
  6           OI.QUANTITY,
  7           OI.PRICE AS ITEM_PRICE,
  8           O.TOTAL_AMOUNT,
  9           P.PAYMENT_METHOD,
 10           P.PAYMENT_STATUS
 11   FROM PURPLE_ORDER O
 12   JOIN PURPLE_CUSTOMER C
 13   ON O.CUSTOMER_ID = C.CUSTOMER_ID
 14   JOIN PURPLE_ORDER_ITEM OI
 15   ON O.ORDER_ID = OI.ORDER_ID
 16   JOIN PURPLE_PRODUCT PR
 17   ON OI.PRODUCT_ID = PR.PRODUCT_ID
 18   LEFT JOIN PURPLE_PAYMENT P
 19   ON O.ORDER_ID = P.ORDER_ID;
```

```
  ORDER_ID ORDER_DAT CUSTOMER_NAME
---------- --------- --------------------------------------------------
CITY
------------------------------
PRODUCT_NAME                                       QUANTITY ITEM_PRICE
-------------------------------------------------- ---------- ----------
TOTAL_AMOUNT PAYMENT_METHOD                 PAYMENT_STATUS
------------ ------------------------------ --------------------
      1001 20-OCT-26 Riya
Chennai
Keyboard                                                1        999
        1498 UPI                            Successful


  ORDER_ID ORDER_DAT CUSTOMER_NAME
---------- --------- --------------------------------------------------
CITY
------------------------------
PRODUCT_NAME                                       QUANTITY ITEM_PRICE
-------------------------------------------------- ---------- ----------
TOTAL_AMOUNT PAYMENT_METHOD                 PAYMENT_STATUS
------------ ------------------------------ --------------------
      1003 22-OCT-26 Riya
Chennai
Keyboard                                                1        999
         999 Debit Card                     Successful


  ORDER_ID ORDER_DAT CUSTOMER_NAME
---------- --------- --------------------------------------------------
CITY
------------------------------
PRODUCT_NAME                                       QUANTITY ITEM_PRICE
-------------------------------------------------- ---------- ----------
TOTAL_AMOUNT PAYMENT_METHOD                 PAYMENT_STATUS
------------ ------------------------------ --------------------
      1001 20-OCT-26 Riya
Chennai
Mouse                                                   1        499
        1498 UPI                            Successful
```

4. Display Customer Purchase History

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_ID,
  4         O.ORDER_DATE,
  5         PR.PRODUCT_NAME,
  6         OI.QUANTITY,
  7         O.TOTAL_AMOUNT
  8  FROM PURPLE_CUSTOMER C
  9  JOIN PURPLE_ORDER O
 10  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 11  JOIN PURPLE_ORDER_ITEM OI
 12  ON O.ORDER_ID = OI.ORDER_ID
 13  JOIN PURPLE_PRODUCT PR
 14  ON OI.PRODUCT_ID = PR.PRODUCT_ID
 15  ORDER BY C.CUSTOMER_NAME;

CUSTOMER_ID CUSTOMER_NAME                                    ORDER_ID
----------- ---------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                      QUANTITY
--------- ---------------------------------------------- ----------
TOTAL_AMOUNT
------------
        102 Bhava                                             1002
21-OCT-26 Headphones                                           1
        1499

        101 Riya                                              1001
20-OCT-26 Mouse                                                1
        1498
```

5. Generate Multi-Table Reports

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         C.CITY,
  4         O.ORDER_ID,
  5         O.ORDER_DATE,
  6         PR.PRODUCT_NAME,
  7         OI.QUANTITY,
  8         OI.PRICE,
  9         O.TOTAL_AMOUNT,
 10         P.PAYMENT_METHOD,
 11         P.PAYMENT_STATUS
 12  FROM PURPLE_CUSTOMER C
 13  JOIN PURPLE_ORDER O
 14  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 15  JOIN PURPLE_ORDER_ITEM OI
 16  ON O.ORDER_ID = OI.ORDER_ID
 17  JOIN PURPLE_PRODUCT PR
 18  ON OI.PRODUCT_ID = PR.PRODUCT_ID
 19  LEFT JOIN PURPLE_PAYMENT P
 20  ON O.ORDER_ID = P.ORDER_ID
 21  ORDER BY O.ORDER_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
CITY                           ORDER_ID ORDER_DAT
------------------------------ --------- ---------
PRODUCT_NAME                                       QUANTITY      PRICE
-------------------------------------------------- ---------- ----------
TOTAL_AMOUNT PAYMENT_METHOD                 PAYMENT_STATUS
------------ ------------------------------ --------------------

         101 Riya
Chennai                             1001 20-OCT-26
Keyboard                                                1        999
         1498 UPI                           Successful
```